

1. Theories of Elastic Failure

2. Transmission Shafts

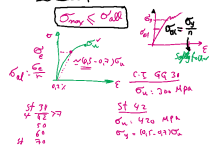
In case of σ_x or $\sigma_y = 2\tau$

$$\sigma_{max/min} = \frac{\sigma_x}{2} \pm \sqrt{\left(\frac{\sigma_x}{2}\right)^2 + \tau^2}$$

$$\tau_{avg} = \sqrt{\left(\frac{\sigma_x}{2}\right)^2 + \tau^2}$$

Theories of Elastic Failure

1. Max. Principal Normal stress



σ ₁ (MPa)	σ ₂ (MPa)
34	340
37	370
42	420
50	500
60	600
70	700

2. Max. pr. shear stress theory

$$\tau_{max} \leq \tau_{all} \quad \left(\tau_{all} = \frac{\sigma_y}{2n} \right)$$

C.I.	σ _y (MPa)
GG18	180
GG22	220
GG30	300

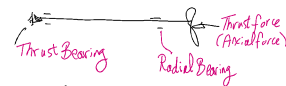
3. Max. strain theory

$$\sigma_{max} - \tau_{min} \leq \sigma_{all}$$

4. Max. pr. strain energy theory

$$\sigma_{max}^2 + \sigma_{min}^2 - 2\tau_{max}\tau_{min} \leq \sigma_{all}^2$$

Transmission Shafts



Solid shafts under Combined Bending & Torsion Only

$$\sigma = \frac{M}{I} y = \frac{32M}{\pi d^3}$$

$$\tau = \frac{T}{J} R = \frac{16T}{\pi d^3}$$

$$\tau_{max} = \frac{16}{\pi d^3} \sqrt{M^2 + T^2} \leq \tau_{all}$$

$$\sigma_{max} = \frac{16}{\pi d^3} \left[M \pm \sqrt{M^2 + T^2} \right] \leq \sigma_{all}$$

$$(K_b) \text{ \& } (K_t)$$

combined fatigue and shock factor
for Bending & Torsion Respectively
 $k_b \rightarrow k_t \Rightarrow 1-3$

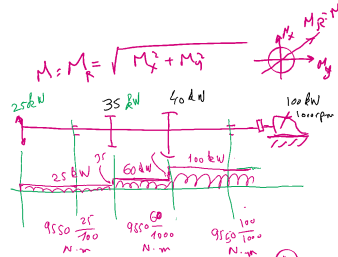
$$\sigma_{max} = \frac{16}{\pi d^3} \left[K_b M \pm \sqrt{(K_b M)^2 + (K_t T)^2} \right] \leq \sigma_{all}$$

$$\tau_{max} = \frac{16}{\pi d^3} \left[\sqrt{(K_b M)^2 + (K_t T)^2} \right] \leq \tau_{all}$$

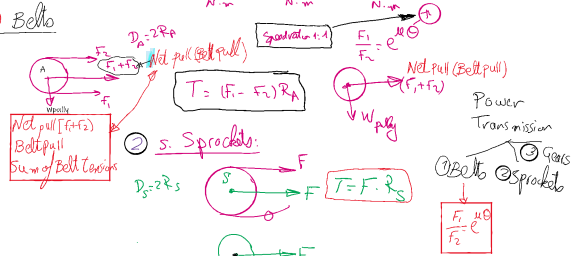
$$T = 9550 \times (\text{power}/N)$$

T>>>>>> N.M
Power>>>> kW
N>>>>>> rpm

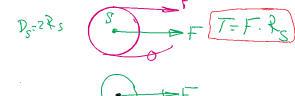
power = 100 kW
N = 1000 rpm
T = 955 Nm



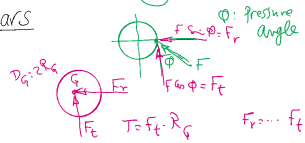
① Belts



② S. Sprockets:



③ GEARS



- ① Draw B.M.D M_x, M_y
- ② Draw T.D.
- ③ Draw N.F.D if found.
- ④ Find the Critical Sec.
- ⑤ $M = \sqrt{M_x^2 + M_y^2}$
- ⑥ Loading Conditions at the critical Sec.

Speed ratio ≥ 1

$\Rightarrow N, F, S, M, T$

$M_b = k_b M$ & $T = k_t T$

$\sigma_{max} = \frac{\sigma}{2} \pm \sqrt{\left(\frac{\sigma}{2}\right)^2 + \tau^2} \leq \sigma_{all}$

$\tau_{max} = \sqrt{\left(\frac{\sigma}{2}\right)^2 + \tau^2} \leq \tau_{all}$ or $\frac{\sigma}{2n}$ or $\frac{\tau}{n}$

$\Rightarrow M$ & T only and Solid cross section:

$\sigma_{max} = \frac{16}{\pi d^3} [k_b M + \sqrt{(k_b M)^2 + (k_t T)^2}] \leq \sigma_{all}$

$\tau_{max} = \frac{16}{\pi d^3} \sqrt{(k_b M)^2 + (k_t T)^2} \leq \tau_{all}$

Solved examples: